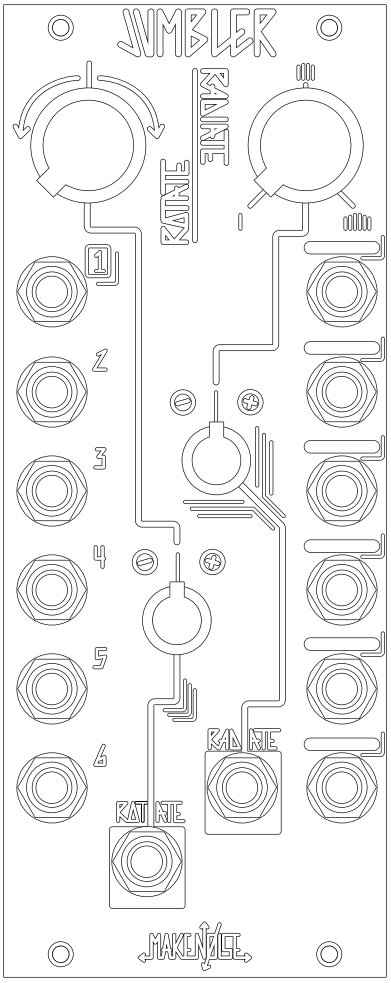
JUMBLER



MAKENOISE

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LIMITED WARRANTY

Make Noise warrants this product to be free of defects in materials or construction for a period of one year from the date of purchase (proof of purchase/invoice required).

Malfunction resulting from wrong power supply voltages, backwards or reversed eurorack bus board cable connection, abuse of the product, removing knobs, changing faceplates, or any other causes determined by Make Noise to be the fault of the user are not covered by this warranty, and normal service rates will apply.

During the warranty period, any defective products will be repaired or replaced, at the option of Make Noise, on a return-to-Make Noise basis with the customer paying the transit cost to Make Noise.

Make Noise implies and accepts no responsibility for harm to person or apparatus caused through operation of this product.

Please contact technical@makenoisemusic.com with any questions, Return To Manufacturer Authorization, or any needs & comments.

http://www.makenoisemusic.com



About This Manual:

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INSTALLATION

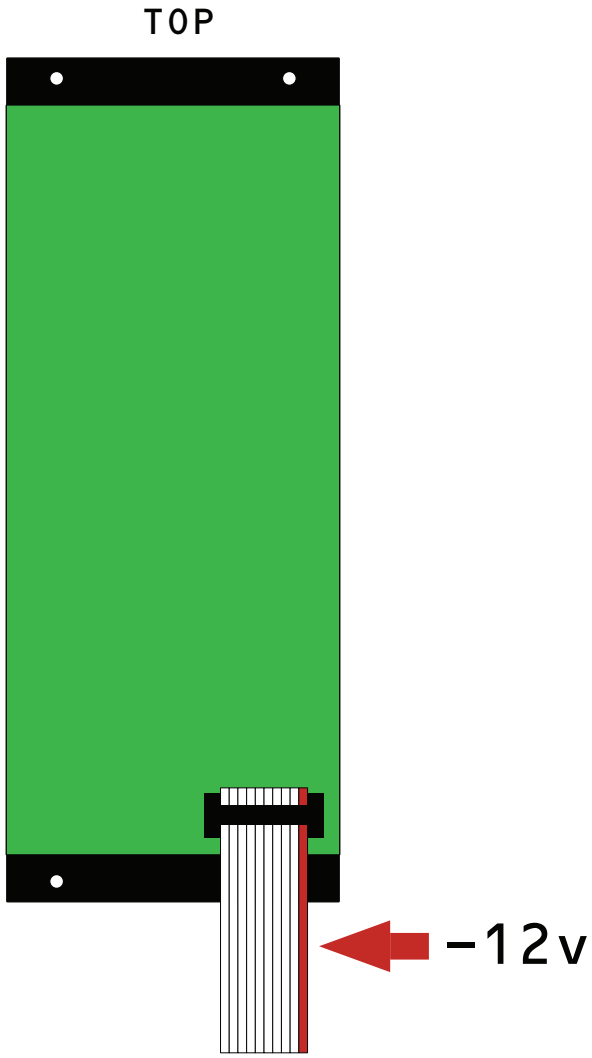
Electrocution hazard!

Always turn the Eurorack case off and unplug the power cord before plugging or unplugging any Eurorack bus board connection cable. Do not touch any electrical terminals when attaching any Eurorack bus board cable.

The Make Noise Jumbler is an electronic music module requiring 178mA of +12VDC and 92mA of -12VDC regulated voltage and a properly formatted distribution receptacle to operate. It must be properly installed into a Eurorack format modular synthesizer system case.

Go to http://www.makenoisemusic.com/ for examples of Eurorack Systems and Cases.

To install, find 10HP in your Eurorack synthesizer case, confirm proper installation of Eurorack bus board connector cable on backside of module (see picture below), plug the bus board connector cable into the Eurorack style bus board, minding the polarity so that the RED stripe on the cable is oriented to the NEGATIVE 12 Volt line on both the module and the bus board. On the Make Noise 6U or 3U Busboard, the negative 12 Volt line is indicated by the white stripe.



Please refer to your case manufacturer's specification for location of the negative supply.

INTRODUCTION

The Make Noise **Jumbler** is a six channel signal combiner and distributor for the **New Universal Synthesizer System** or any Eurorack modular synthesizer.

Jumbler gives you simple two-parameter access to the rearrangement of origins and destinations for up to six modular signals. One to six inputs are routed to one to six outputs, in an order determined by the **Rotate** control. Signals can be isolated or blended together using the **Radiate** control.

Within the field of Jumbler uses there are three primary endpoints/singularities that constitute "typical uses:"

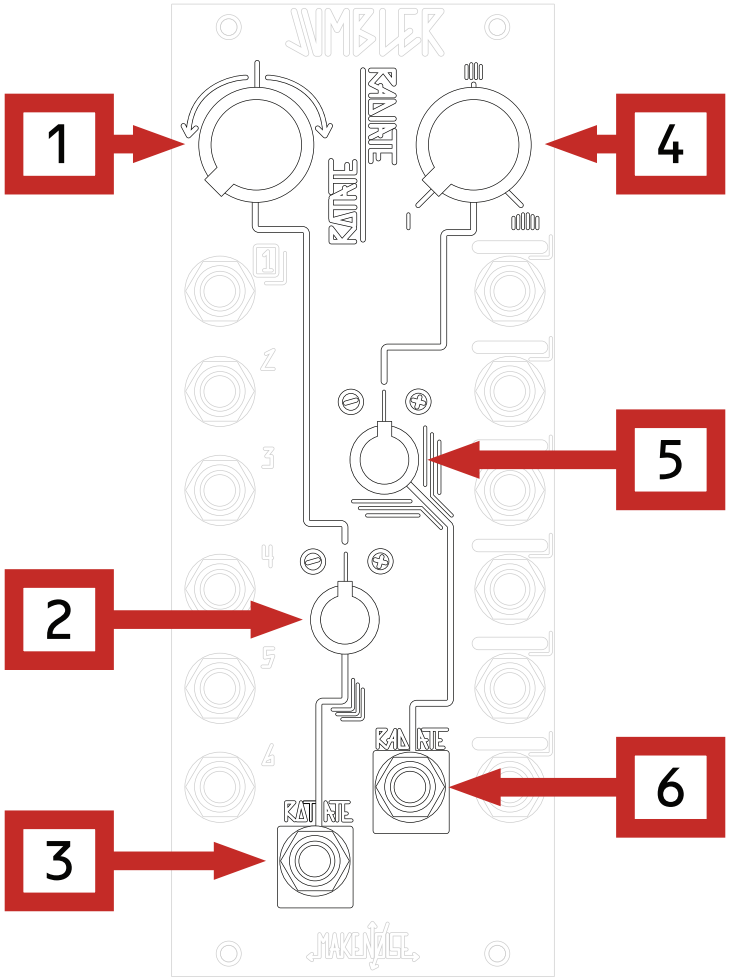
- Signal distributing. Send a single origin signal to any of the six destinations, or to more than one (up to all six).
- Signal funneling. Send six origin signals to a single destination, either one by one, or mixed.
- Jumbling. Send six origin signals to six different destinations, rotating the routing and blending at will.

1:6, **6:1**,and **6:6** are the "endpoints" but of course any combination of ins and outs can be used.

The primary patch that gives Jumbler its name is a universal insert into any modular patch. While building a patch, set the Jumbler controls to their initial values, choose a particular type of signal (CV, audio, or gate) and each time you patch that type of signal somewhere in the system, patch it in series through an input and output of Jumbler. Once you have all six channels patched through, use the Rotate and Radiate controls to Jumble the patch, creating any number of routing variations that can be modulated, sequenced etc. Different signal types may also be Jumbled for on-the-fly shakeups of the entire musico-conceptual character of a patch.

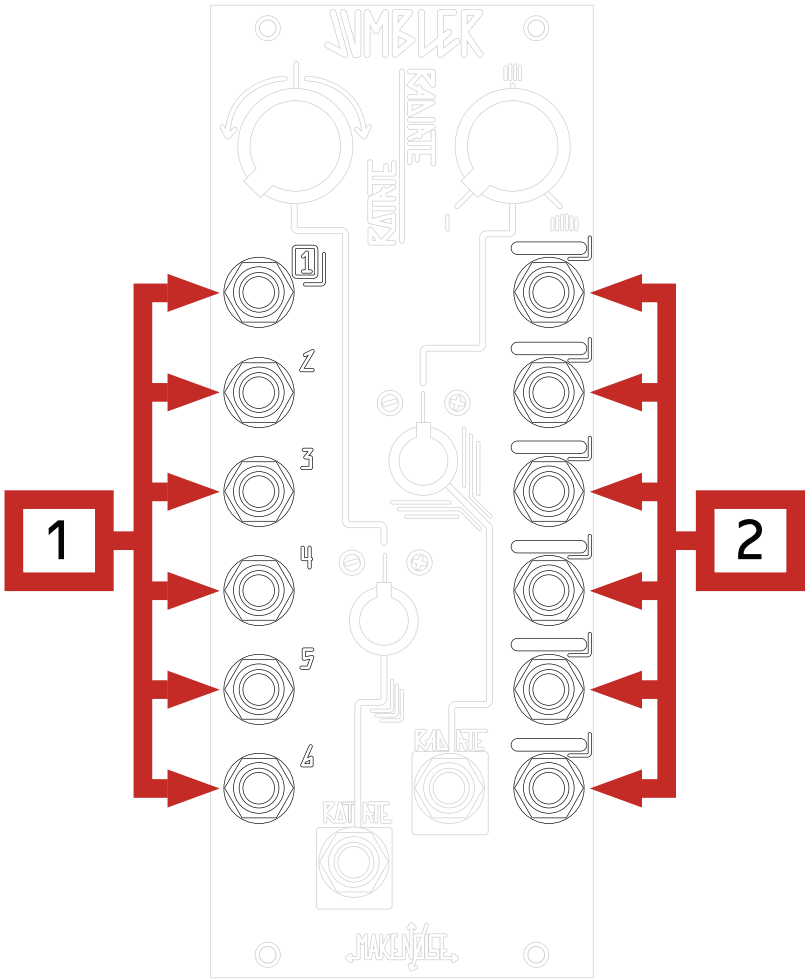
Jumbler is a great module for gate or clock combinations, hybrid control voltage mashing, waveform mixing, on-the-fly effects routing change-ups, and more.

PANEL CONTROLS



1. **Rotate Panel Control:** Bi-polar control for Rotate. Selects basic routing of inputs to outputs.
2. **Rotate CV Attenuverter:** Bi-polar CV input attenuverter.
3. **Rotate CV Input**
4. **Radiate Panel Control:** Sets the amount of radiation from each output into adjacent outputs.
5. **Radiate CV Attenuverter:** Bi-polar CV input attenuverter.
6. **Radiate CV Input**

PANEL CONTROLS



1. **Signal Inputs:** Six signal Inputs.

2. **Signal Outputs:** Six signal Outputs. Signals are sent from inputs to outputs according to the Rotate and Radiate controls.

ROTATE, RADIATE, AND CHANNEL OUTS

Rotate

The Rotate panel control selects the basic routing of inputs to outputs. At 12:00, Input 1 is sent to Output 1, Input 2 to Output 2, etc. Turning it slightly Clockwise Rotates the outputs:

1 6
2 1
3 2
4 3
5 4
6 5

Slightly further:

1 5
2 6
3 1
4 2
5 3
6 4

And so forth. At full clockwise, input 1 goes "beyond" output 6 and all outputs turn OFF. Rotating counterclockwise from 12:00 rotates the outputs in the opposite direction. The output that currently receives its signal from Input 1 is always indicated in BLUE.

Radiate

The Radiate panel control sets the amount of radiation from each output into adjacent outputs. At full CCW, each input signal occupies its own space in the Rotation. As Radiate is turned up, any signal at a given output is blended into the "next" and "previous" outputs. For example if there is a signal in Output 3, turning up Radiate begins to add this signal to Outputs 2 and 4. (Outputs 1 and 6 are also considered "adjacent" to each other by Radiate.) With Radiation present, Rotation results in a smoother transition from one set of signals to the next at a given output. Turning up Radiate further adds additional outputs, until at full CW each input's signal is present in every output.

Rotate takes place "prior" to Radiate. For example if you patch a signal to Input 1, Rotate it so that it appears in Output 3, and Radiate slightly, it will appear in outputs 2, 3, and 4.

Channel Outputs

Signals are sent from inputs to outputs according to the Rotate and Radiate controls. Output LEDs indicate:

Blue: The output that is currently receiving from Input 1 according to the Rotate control
Green/Red: Positive or negative signal indicator displaying current activity at each output.

With Rotate at 12:00 and Radiate full counterclockwise, signals pass through directly from each numbered input to the output directly to its right.

TIPS & TRICKS

- Sequence Rotate and/or Radiate CV to dial in specific "Jumblings" that you can step through at will (use a Pressure Points or 0-CTRL to select them manually!)

- All signals can be turned OFF with Rotate set to extremes. In this case you can use the Rotate CV input as a "scanning VCA" to turn multiple signals on and off via different envelope shapes.

- No need to use all the inputs and outputs every time! Patch signals to non-consecutive inputs to create purposeful "gaps" across outputs when Rotating, or use Radiate to momentarily bridge these gaps.

PATCH CORNER

Arpeggio/Chord Creation

Patch up to 6 individual tones to Jumbler's inputs, tuned to be notes of a chord. Modulate Rotate to create an arpeggio-ramps for linear arpeggios, rounded shapes for forward/reverse, random for randomized notes. Modulate Radiate to blend these notes into chords, with larger amounts of modulation resulting in chords with more notes. Use Rotate and Radiate together to shift the emphasis of the chord toward certain clusters of notes in the collection.

Mono/Stereo/Quad/etc. Scanning Mixer

- Patch any number of inputs to Jumbler and patch one of its outputs to the monitors. Use Rotate and Radiate to create a mix of the inputs at the output.
- Same as above, but patch two of its outputs (say, 1 and 4) to the monitors and pan these outs in hard stereo.
- Same as above, but patch four of its outputs (dealer's choice) to the monitors of a quad system.
- Same as above, but take an odd number of its outputs to the monitors of a unique 3:1, 5:1, etc. speaker arrangement

In all of these examples, modulation of Rotate is key for creating spatial animation.

JUMBLE

Set Jumbler. such that Rotate is at noon and Radiate is CCW. Patch your modulation signals through Jumbler, such that a function modulating the CV Input of a LPG passes first through Jumbler (say, patch the function to the Channel 1 input and then patch the Channel 1 output to the LPG). Do this for all of the modulation signals in a patch (or up to 6 of them). Then, once complete, modulate Rotate to Jumble the patch! Return Rotate to noon to reset to your initial position.

Variable FX Send

Patch a signal to a Jumbler input and patch any number of Jumbler outputs to the inputs of signal processing modules (delays, reverbs, filters, etc.), all set fully Wet. Mix the outputs of the processing modules together and monitor the output of the mixer. Modulating Rotate routes the signal to the signal processing modules, increasing Radiate routes the signal to more than one at once.